

Note on the COM/orientation potential created by Gaussian optical tweezers

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1 Centre of Mass

Optical tweezers trapping potential is usually modeled as proportional to the light intensity profile (a Gaussian beam), which near the focus is well-approximated as harmonic.

1.1 Full Gaussian Potential (radially symmetric)

Here is the full potential expression:

$$U(r, z) = -U_0 \frac{w_0^2}{w(z)^2} \exp\left(-\frac{2r^2}{w(z)^2}\right) \quad (1)$$

where $w(z) = w_0 \sqrt{1 + (z/z_R)^2}$ is the beam waist, $z_R = \pi w_0^2/\lambda$ is the Rayleigh range and U_0 is the trap depth, which in Rayleigh approximation, (valid for a particle small compared to λ) is this:

$$U_0 = \frac{2\alpha}{\varepsilon_0 c \pi w_0^2} P \quad (2)$$

where α is the particle's polarizability (can be obtained from Clausius-Mossotti), P is the laser power.

1.2 Harmonic Approximation Near the Trap Center

Valid for small displacements:

$$U(x, y, z) \approx \frac{1}{2}k_x x^2 + \frac{1}{2}k_y y^2 + \frac{1}{2}k_z z^2 \quad (3)$$

with trap stiffnesses k_x, k_y, k_z that can be obtained from Taylor expanding $U(r, z)$ to the second order around the origin.

If you use this, the particle cannot escape from the trap (potential depth is infinite).

1.3 Note about the particle size

The potential expression is strictly only true for a Rayleigh particle (smaller than a wavelength). Perhaps in the future, once we settle on the particle shape, we can replace this with a Mie- regime expression (comparable to wavelength). This will have to be obtained numerically.

2 Orientation

In this section we assume a cylindrically-symmetric particle, e.g. a rod with fixed aspect ratio $p = L/D$ (length/diameter), orientation θ (angle from the z -axis) is set by two competing torques, both well approximated as sinusoidal.

2.1 Intensity gradient torques

When L is comparable to w_0 , tilting the rod pushes part of its length into lower-intensity regions, independent of any polarizability anisotropy. In this section we assume a scalar polarizability α . **The potential is then proportional to optical intensity of the Gaussian beam**, integrated over the particle volume, sampling the local field wherever the particle sits:

$$U_{\text{grad}} = -\frac{\alpha_v}{2} \int_V |E|^2(\mathbf{r}) dV \quad (4)$$

where α_v is polarizability per unit volume. Since α_v is constant, orientation θ enters *only* through which region of space the fixed-shape volume occupies as the rod tilts — a purely geometric effect.

We can now calculate the potential depth $\Delta U_{\text{grad}} = U_{\text{grad}}(\theta = 0) - U_{\text{grad}}(\theta = \pi/2)$, just by reorienting the integration volume in Eq. (4).

Assuming the potential is sinusoidal (which is fair, I think), we can match to

$$U_{\text{grad}}(\theta) \approx -\frac{\Delta U_{\text{grad}}}{2} \cos(2\theta) \quad (5)$$

and get the torque which tries to align the particle with z -axis:

$$\tau_{\text{grad}}(\theta) = -\frac{dU_{\text{grad}}}{d\theta} \approx -\kappa_{\text{grad}} \sin(2\theta) \quad (6)$$

From a quick back-of-the-envelope, the torque (κ_{grad}) should scale like $V(\frac{L}{w_0})^2$, where V is the particle volume. Torque vector direction is always perpendicular to z and to the particle long axis.

2.2 Polarization Torque

Anisotropic polarizability happens because of the particle elongated shape. The particle is the most polarizable along its long axis ($\alpha_{\parallel} > \alpha_{\perp}$) and favors alignment of the long axis with the field polarization direction ($\phi = 0$ measured from the polarization axis):

$$U_{\text{pol}}(\phi) \approx -\frac{\kappa_{\text{pol}}}{2} \cos(2\phi) \quad (7)$$

$$\tau_{\text{pol}}(\phi) = -\frac{dU_{\text{pol}}}{d\phi} \approx -\kappa_{\text{pol}} \sin(2\phi) \quad (8)$$

with

$$\kappa_{\text{pol}} = \frac{1}{2}(\alpha_{\parallel} - \alpha_{\perp})E_0^2. \quad (9)$$

Note that τ_{pol} is not independent on θ , just because the polarization torque disappears if particle points along z ($\theta = 0$) and is maximum for $\theta = \pi/2$. The angles we use θ and ϕ are not azimuthal and polar Euler angles (ϕ is defined differently). **Torque direction is always perpendicular to the polarization and to the particle long axis.**

Since the two torques pull toward orthogonal directions (z -axis vs. transverse polarization), the net equilibrium orientation depends on the relative size of κ_{grad} and κ_{pol} , i.e. on L/w_0 and aspect ratio p .

3 Couplings

I think the best way to tackle this is to numerically calculate forces for different orientations as well as torques for different positions.

This Github code can do this easily: <https://github.com/barra9408/info-patterns-repo>. However, below I try to roughly explain where the coupling comes from.

3.1 Orientation affecting the COM

I think there is two components to this. First, we have assumed only peak intensity matters, and used its relationship to tweezers power to obtain the potential depth U_0 . However, the effective optical intensity "seen" by an extended particle might be better approximated by an average optical intensity over the particle volume. That will depend on the orientation of the particle.

Second, the polarizability is actually anisotropic (tensor), not a scalar quantity, just because the particle is elongated. Therefore, the polarizability will depend on the orientation. Expressions exist for polarizability of an ellipsoid in a Rayleigh regime, but in our case we can just crunch this numerically (since we have inhomogenous field).

3.2 COM affecting the orientation

When integrating the intensity to calculate the gradient torque (or taking the field amplitude for polarization-induced torque), we assumed that the particle sits in the origin. However, as soon as the particle moves away from the origin, the intensity sampled by particle will change. That means that the orientation potential (and torques acting on the particle) will depend on the particle's position.

4 Cases we might want to include in the algae study

- (a): Simplest case, harmonic COM potential from Eq. (3) and gradient torques from Eq. (5).
- (b): Another interesting case would be full COM potential from Eq. (1) plus gradient torques from Eq. (5). There could be an interesting escape-from-trap dynamics. Possibly this is already enough.
- (c): Not sure if the polarization is worth including (I doubt it), but in principle we could also try to estimate the polarization effect by doing harmonic COM potential from Eq. (3) plus gradient torques Eq. (5) and polarization torques Eq. (8).
- (d): Full numeric forces and torques with all the cross couplings.